

A Machine Reads Grothendieck’s Working Notes: Transcribing and Encoding Mathematical Manuscripts

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Abstract

This article reports the transcription by a large language model of the working notes of the mathematician Alexandre Grothendieck (1928–2014), held by the University of Montpellier: 178 openly accessible folders, 16,074 digitised pages, 1949 to 1991, in a hand that ran faster than it could be read and a notation of his own. The model transcribed the pages twenty at a time into a restricted LaTeX of nine apparatus macros; a mechanical converter derives from each file a TEI document in which the mathematics is carried as TeX inside `<formula>` and commutative diagrams as `tikz-cd` inside `<figure>`; a second register, a modernised reading in current notation, is derived from the transcription and never from the hand. The edition stores no page of the fonds: the facsimile beside each reading is Montpellier’s own file, streamed through a relay as it is looked at. By 3 October 2026, 586 passes had read 8,565 pages in 160 folders, and the apparatus records 64,030 words left illegible, 7.5 per cent of the words read. The article sets out the encoding and its reasons, the derived registers of names, dates, letters and hands that the transcription makes possible, and the limits: no batch has been checked by a person, only some hundred and ten points on the facsimile, one macro was found carrying two senses, and TEI’s formula element carries mathematics it does not describe. The use of generative AI is declared.

Keywords. mathematical manuscripts; large language models; transcription; critical apparatus; mathematical notation; archival rights; Alexandre Grothendieck.

1 Introduction

Alexandre Grothendieck handed his mathematical papers to his former student Jean Malgoire in two parts, in the summer of 1990 and on 28 July 1995; Malgoire kept them at home until 2010, deposited them at the University of Montpellier, and gave them to it in 2012 (Université de Montpellier n.d.). They form the fonds that the university catalogued in 2015 and 2016 and has digitised in part: 178 folders in open access, 16,074 pages by the inventory’s count, arranged in twenty-two groups in the order he filed them (Université de Montpellier n.d.). They are working notes. He wrote for himself, day by day, in ink and pencil, over and between his own lines; he took the same chapter up four times; he abandoned a proof and picked it up thirty pages later; he wrote on the backs of administrative letters, of seminar notes and of other people’s typescripts. Some of it has been transcribed by hand, by mathematicians who gave years to the work, and that work can be measured against the inventory. It covers twenty folders and 5,129 pages, a third of the pages in open access, and five works: *Les Dérivateurs*, transcribed whole into LaTeX (Künzer, Malgoire and Maltsiniotis n.d.; five folders,

2,135 pages), the *Esquisse d'un programme* (Grothendieck 1997; one folder, 313 pages), and three works edited in part — the first thirty-seven sections of the *Longue Marche à travers la théorie de Galois* (Grothendieck 1995; four folders, 1,584 pages), the first volume of *À la poursuite des champs* (Maltsiniotis 2022; eight folders, 837 pages), and an English translation of the notes for EGA V (Grothendieck Circle n.d.; two folders, 260 pages). The correspondence with Serre (Colmez and Serre 2001) edits letters of which the folders hold a few. What he printed — EGA in the *Publications mathématiques de l'IHÉS*, now on Numdam (Grothendieck and Dieudonné 1960–1967), the SGA and the FGA — is the text some folders draft, and not a transcription of them. The other 158 folders, 10,945 pages, had no transcription at all, and most of their pages had not been read by anyone since they were written.

This article reports an experiment in reading it with a large language model as the first transcriber, under constraints designed to make its reading checkable rather than trustworthy. It is the second report of a method first run on the ledgers of Edward and Josephine Hopper (Hua 2026), and it is written for what this corpus changes. Three things change. The document is mathematics, and mathematics is a notation before it is a language, so the question of how a transcription encodes it — and how TEI carries it — is the question the Hopper ledgers never raised. The corpus is two orders of magnitude larger, which turns the human check from a debt into the main fact about the edition. And the witness cannot be copied: the fonds is under copyright, a third of its pages may not be circulated at all, and the edition holds none of it, so the reading and its facsimile meet only in the reader's browser.

The objective is methodological. No claim about Grothendieck's mathematics is made here that the demonstration would need to be correct for. The article presents the fonds (Section 3), the transcription method and its rules (Section 4), the two registers (Section 5), the TEI encoding and what it does with mathematics (Section 6), the interface and the derived registers (Section 7), the findings and how they are kept from becoming claims (Section 8), and the limits (Section 9). Four words are kept apart, as in the first report: a reading is *reproducible* when the process that made it can be rerun, *auditable* when a reader can find the evidence for each sentence, *reliable* when like cases are treated alike, and *valid* when it is right. This edition claims the first three and not the fourth.

2 Related work

The transcription of handwriting by large language models is recent and moving. Humphries and colleagues (2025) report character error rates of about six to seven per cent for general-purpose models on eighteenth- and nineteenth-century English manuscripts; Crosilla, Klic and Colavizza (2025) benchmark several models and discuss when a model's linguistic prior helps and when it hinders. Mathematics is the extreme case of the second. A model that has read a great deal of mathematics will complete a formula as readily as it completes a sentence, and a completed formula is the more dangerous of the two, because it is well-formed: an index that makes an identity true is not a reading of the index on the page. The rule this edition sets against that prior — nothing illegible is guessed, even where the mathematics would supply it — is the one that shapes everything else, and Section 9 measures what it costs.

The evaluation of models as producers of TEI is more recent still. Strutz (2026) proposes a framework for assessing model-generated TEI from syntactic validity to semantic accuracy. As in the first report, the model here writes no TEI at all: it writes a restricted language whose macros correspond one to one with TEI constructs, and the TEI is derived. The precedent for

a reduced intermediate language converted up into conformant TEI is Cummings and Willcox (2013), whose `tei_corset` bounded what a keying company could write; these macros bound what a model may write, and the consequence is shared, that the converter must be something a test can hold to account.

Mathematics in TEI has a short answer in the Guidelines and a long one outside them. The Guidelines provide `<formula>`, whose `@notation` names the notation the content is in, and leave that content to another vocabulary: MathML, TeX, or an image (TEI Consortium 2026). MathML is the W3C's vocabulary for mathematical structure and presentation (W3C 2014), and a TEI file may embed it. Section 6 explains why this edition carries TeX instead, and what that choice gives up. Hand attribution, which the first report treated at length, is a lesser question here: the fonds is in one hand, his, except where it holds other people's letters and typescripts, and those are marked by what they are rather than by their letterforms (Stokes 2015).

3 The fonds

3.1 The inventory, and the three numberings

The inventory of the fonds is complete and rigorous, and was made for an archive and not for a reader. Each folder has a shelfmark (19, 140–2, 156–9), a title — his, pencilled on the cover, or the archivists', in square brackets where the folder bore none — a dating, and a page count; the folders stand in twenty-two groups, from *Cristaux* and *Théorie des motifs* to the *Longue marche à travers la théorie de Galois* and *Dérivateurs*. Each folder is served as one PDF, between 4 and 270 megabytes.

Three numberings overlap on every page. Grothendieck's own, where he paginated: the Long March runs from 1 to 787 across four boxes. The archivists', pencilled at the foot of every page, which is what the inventory counts and what the edition calls a page. And the PDF's, one ahead of the archivists', because the university's file opens on a generated cover. The edition addresses a page by the archivists' number and converts to the PDF's only at the last moment, when it asks Montpellier for a page; a reading that cited the PDF's number would cite a sheet the archivists never numbered.

3.2 What the pages are

The pages are drafts, and the drafting is the object. A restart is not a repetition to delete. A verso is often unrelated to its recto — an administrative letter, a typescript returned by a secretary — and was digitised deliberately, sometimes upside down. The dates are nearly all inferred. Figure 1 sets the inventory's datings against the dates the transcriptions have found written on the leaves: of the 153 folders the inventory dates, 111 rest in part on a deduction the archivists mark with their square brackets, and 25 more are « s.d. », undated; the transcriptions have found 198 dates on the leaves themselves, in 64 folders, 47 of them machine stamps on the computer listings whose backs he wrote on in the 1980s, which date the paper and not the writing; a reading of the facsimiles of eight edited folders adds 40 more, eight of them stamps.

3.3 What may be read, and what was

Of the fonds' some 28,000 pages, about 18,000 may be circulated; third-party correspondence may not, without permission. The 178 folders in open access hold 16,074 pages, which at

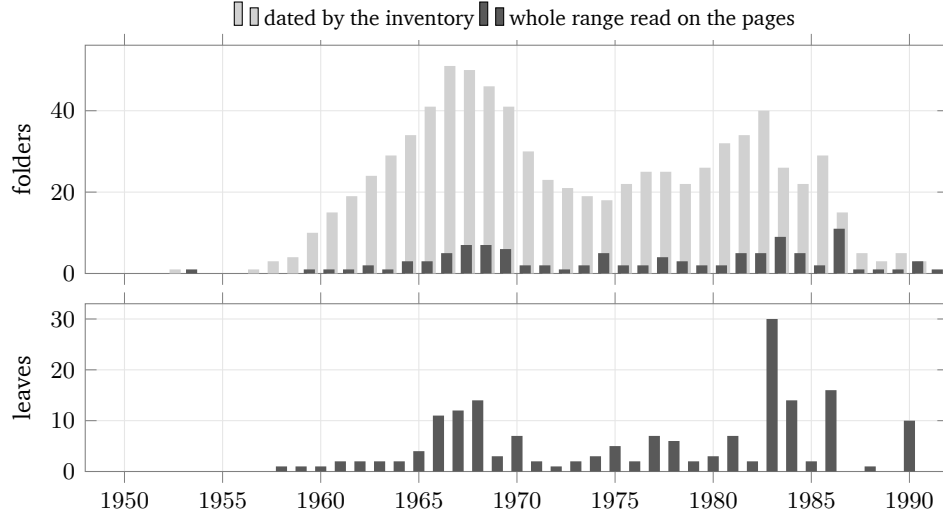


Figure 1: Two kinds of date on one axis, each on its own scale. Above: for each year, how many of the 153 dated folders the inventory’s ranges cover (an open end, « à partir de 1982 », counted five years on), and, darker, how many of them rest on a range the archivists read on the pages rather than deduced in their square brackets; 25 folders are « s.d. ». Below: the 177 dates the transcriptions have found written on the leaves themselves, machine stamps on reused paper left out. From `src/content/catalogue.ts` and `src/content/dated-leaves.json` (2026-10-03), after the site’s page *Timeline*.

twenty pages to a batch make 884 batches. Twenty folders — 5,129 pages, 266 batches — are already edited by mathematicians, among them the *Dérivateurs*, the Long March, *Pursuing Stacks* and the *Esquisse d’un programme*, and the edition marks them and links to the edition rather than read them again, since a machine’s first pass beside a scholar’s edition is waste. Table 1 gives the state of the reading on 3 October 2026.

Table 1: The fonds as counted on 3 October 2026: the inventory, the part already edited by others and set aside, and what the passes have read. A batch is twenty of the archivists’ pages; a transcribed page is one that carries mathematics, the others being skipped by rule.

	Folders	Pages	Batches
Open access (inventory)	178	16,074	884
Already edited, set aside	20	5,129	266
Left to this edition	158	10,945	618
Begun	154		
Transcribed		8,565	573
Read whole and modernised	91		256

4 The transcription method

4.1 The schema, and its values here

Figure 2 gives the arrangement, which is the one the first report describes and which this edition was in fact the first to run: a *witness* held by the institution and never committed; a *reading*, the one file that is written, by the model first and by a person thereafter, in a language whose every construct is declared; *derivations*, computed from the reading by scripts on every build; *verifications*, whatever a machine can check of them; and an *address*, the reading placed

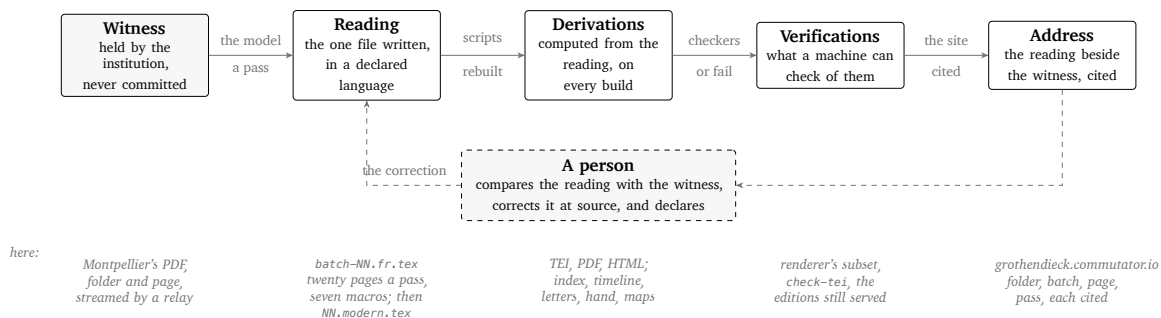


Figure 2: The schema this edition shares with its sibling on the Hopper ledgers (Hua 2026), and its values in this one. Five stations and one loop: the model turns a witness into a reading, in passes whose header names the model and the date; scripts derive everything else from the reading and regenerate it on every build; checkers fail the build on what a machine can check; the site places the reading beside the witness and cites both by reference. The dashed loop is the person, who compares the reading with the witness and corrects the reading, which alone changes anything downstream. Four invariants hold at every station: the witness is never committed; only the reading is written, everything else is derived and compared; every assertion carries an address into the witness; and the person's verdict is declared, not observed. The grey row gives this edition's values. Figure by the author.

beside its witness and cited by the holder's reference and the edition's dated pass. The loop is the person. What distinguishes this edition from the ledgers' is only the values the stations take, and three of them are new: a model that reads and then restates, a witness that cannot be mirrored for the reader, and a notation that is not text.

4.2 The unit of work and the model

A batch is twenty of the archivists' pages, read in one fresh model context and written as a single file, transcripts/<folder>/batch-NN.fr.tex. The ceiling is not arbitrary. Past twenty pages of this hand, the reading degrades towards the end of the pass with nothing in the output to signal it, and a transcription whose weakening point is unknown cannot be used. The skill that performs the reading admits named models and refuses any other, and each file's header records the model and the date in a fixed form, % Pass: Opus 5.5 (claude-opus-5-5), <date> --- first pass, unchecked against the pages by a human, so that provenance is a fact about the file and not about the session. Of the 586 first passes, 453 were read by Claude Opus 5.5, 102 by Claude Opus 5, 26 by Claude Fable 5.1 and 5 by Claude Fable 5; the Fable models were withdrawn for new passes on 22 September 2026, and the files they read stay as they are, their headers saying so. Thirty-three files in twenty folders carry a later revision below the first pass, dated and attributed in the same way; seven of them apply readings the editor settled on the facsimile in October, and say so, and none makes its batch checked. The batches of a folder may run in parallel, one model context each; the price is continuity, since no batch can read the one before it, and each header says where a run crosses its boundary.

Two aids are handed to every pass, and neither is allowed to settle a reading. The first is a set of tiles: each page cut into six overlapping crops at 700 dpi, with the margin turned upright, because choosing crop rectangles by hand cost one early batch some ninety commands for sixteen pages, and choosing rectangles is not reading. The second is a lexicon of his vocabulary, abbreviations and notation, counted over every transcription so far and regenerated after each run, which keeps apart the words read plainly, the words read with doubt, and the words that occur only in doubt. A pass reading a hard leaf is rarely choosing between a word and no word; it is choosing between candidates, and the candidate that belongs to this material is the

one the lexicon shows. It is built from unverified transcriptions without circularity, because it makes no claim about ink: that *principal* is common in these pages is settled by hundreds of plain occurrences, and whether one stroke reads *principaux* is left to the page.

4.3 The rules

The governing rule is the ledgers’ rule translated: an invented word in a letter is a bad reading; an invented index in a formula is a theorem nobody proved. Nothing illegible is guessed, and the mark for an illegible word is never replaced by what the mathematics would require. A reading offered with doubt is marked doubtful. His language stays his — his spelling, his punctuation, his long sentences, his slips, flagged and not corrected — and so does the transcriber’s, since every note is in French, so that the edition a French reader reads straight through has no seam in it.

The transcription is not diplomatic, and says so. The archive’s material apparatus — which verso carried which letter, where a separator sheet fell, which page was scanned upside down — is not its subject; a page that carries no mathematics is simply absent, and the gap in the page numbers is the record. Where a page is too crowded to render whole, the mathematics is transcribed and the surrounding prose compressed, with a note. Commutative diagrams go into `tikz-cd`, never into prose, because that is the notation the mathematics of this fonds is now read in. His own prose stays whenever it is about the mathematics, and it often is: where he stops to say why a definition is the right one, that is the content. What he struck out stays, marked, because a deletion shows where the thought turned.

Three rules concern other people. A typescript or a *mémoire* by someone else is given as a note per page and his marks on it in full, not transcribed; a letter to him is transcribed in full where its writer is a public figure and is otherwise summarised, its writer unnamed; and letters already published in the correspondence with Serre stay condensed to their statements and a summary. Each rule answers a right that the edition does not hold.

4.4 What the model writes

The model writes a restricted LaTeX. Its apparatus is nine macros (Table 2); the mathematics is ordinary LaTeX mathematics, inline and displayed, with the environments a working mathematician uses and `tikz-cd` for diagrams. Six declarations open each file — the shelfmark, the batch, its pages, the inventory’s dating copied verbatim with its brackets, the folder’s title, and a watermark reading *Édition de démonstration* that the PDF prints across every page, because a file without it would claim by silence a status it does not have.

The subset is policed twice, by the renderer that turns each file into the site’s reading view and by the converter to TEI (Section 6), and both refusals now stop the build. An unknown environment is refused by both, naming the file; the converter also refuses a control sequence it would otherwise leave in the TEI as text, and a header line recording a pass that it cannot read. Until late September 2026 a refusal was a warning: the script exited cleanly, the deploy went on, and the refused batch was simply missing from the site. The count of leaked control sequences, which found some thirty files carrying them earlier that month, all corrected at source, is now that refusal rather than a check run on request. Of three failures recorded in the repository as passing the renderer silently, one is now caught: the converter no longer maps `\underline`, so the macro in running prose is a control sequence left as text, and refused. The other two — a backslash-newline inside inline mathematics, and a page-range declaration without a blank line before it — still pass both scripts, and are found by reading the output.

Table 2: The apparatus macros, as the transcription skill declares them, and the TEI elements the converter maps them to. The mathematics itself is carried as TeX (Section 6).

Macro	Meaning	TEI
<code>\page{47}</code>	Page 47 begins, in the archivists' numbering	pb with @n, @facs
<code>\ill{}</code>	Illegible; never guessed	gap reason="illegible"
<code>\uncertain{...}</code>	A reading offered and flagged	unclear
<code>\supplied{...}</code>	What the transcriber supplies and the page does not carry: a missing letter, a clipped word-end	supplied resp="#pass"
<code>\add{...}</code>	His own insertion: above the line, over a struck word, his ink on a typescript	add
<code>\struck{...}</code>	Struck out by Grothendieck	del
<code>\note{...}</code>	The transcriber's note, about the mathematics	note type="editorial"
<code>\drawing{...}</code>	A drawing of his, described rather than redrawn	note type="drawing"
<code>\marginal{...}</code>	A marginal note of his	note type="authorial" place="margin"

4.5 A transcribed page

The following is page 8 of folder 19, *Topos*, from the first batch the edition transcribed of it, as the converter exports it (the page is folder 19, batch 1). The mathematics is TeX inside `<formula>`; the diagram is a figure; the note is the transcriber's and the marginal his; the struck φ and what replaced it are both kept.

```
<pb n="8" facs="https://grothendieck.umontpellier.fr/19.pdf#page=9"/>
<p>qui s'expliciteront en termes des <formula notation="TeX">\varphi_{ji} = f_{jj} g_{ji}</formula>,
    [...]
```

$$\lambda_{kji} : \varphi_{ki} \rightarrow \varphi_{kj} \circ \varphi_{ji} \tag{3.10}$$

```
</formula></p>
<p><figure type="diagram"><formula notation="tikz-cd">\begin{tikzcd}[column sep=small, row sep=
    small]
\varphi_{ki} \arrow[r, "\lambda_{kji}"] \arrow[d, no head, "\parallel" description]
& \varphi_{kj} \varphi_{ji} \arrow[d, no head, "\parallel" description] \\
f_{kj} g_{ji} & f_{kj} \backslash, g_{jj} f_{jj} \backslash, g_{ji}
\end{tikzcd}</formula></figure></p>
<p><note type="editorial" resp="#pass">la flèche provient du morphisme d'adjonction
<formula notation="TeX">\mathrm{id} \rightarrow g_{jj} f_{jj}</formula>, que l'auteur indique par une
flèche verticale sous <formula notation="TeX">g_{jj} f_{jj}</formula></note></p>
<p><note type="authorial" place="margin">pour <formula notation="TeX">i = j = k</formula>,
<formula notation="TeX">\lambda_{kji}</formula> n'est autre que
<formula notation="TeX">\lambda_{ji}</formula></note></p>
<p>[Le cas le plus intéressant est celui où les <del><formula notation="TeX">\varphi</formula></del>
    del>
<add><formula notation="TeX">\alpha_i</formula></add> sont des
isomorphismes (i.e. les <formula notation="TeX">g_{ji}</formula> pleinement fidèles). [...]</p>
```

The last paragraph shows the distinction Section 9.2 takes up. The α_i is his, written over the φ he struck, and is encoded as `<add>`; until 26 September 2026 the converter, following the one macro's declared sense, encoded it as `<supplied>` and credited it to the transcriber.

5 Two registers

The ledgers of the first report needed one text. This fonds needs two, and the reason is the notation. A price of 1927 means what it meant; a $\mathcal{C}$ of 1962 is not drawn like one of 1986, a sheaf of the Séminaire is not named as a sheaf is named today, and a reader who meets «fppf» spelled out on one page and reduced to initials on the next needs to be told it is the same topology. So every folder read whole yields a second document, the *modernised reading*: a résumé written for someone who has not met the subject, then the mathematics in current notation and current names, with footnotes carrying everything the transcription's apparatus carried.

Three decisions govern it. The modernised reading is derived from the transcription and never from the hand: two independent readings of one hand would diverge, and nothing would say which was right. It is made per folder and not per batch, because its job is to make an argument run continuously and his arguments do not stop where the batches do. And it is the one document allowed to depart from the page, which is why it is held to a different standard from the transcription: not *faithful* but *correct as it stands*. Where the manuscript is loose, the reading states what is true and footnotes what the page has. The standard is a reasoning standard, calibrated against one reference reading and stated in the skill as four failure modes to check: a statement strengthened beyond what the page proves, a hypothesis dropped, a notation modernised into a different object, and a gap closed silently.

The résumé closes on a line of English keywords, the modern vocabulary to search under. They are the one piece of either register that is written for search rather than for reading, and the edition treats them as such: they are what the site's search, its index of subjects and its map of the mathematics are built from (Section 7.3), and they are said to be the reading's and not the page's. On 3 October 2026 ninety-one folders had been read whole in this way, carrying 1,903 keywords.

6 The TEI encoding

6.1 Derived, not written

Every transcription is exported to TEI P5 by a script that reads the .tex and nothing else — not the facsimile, not the rendered HTML. The mapping is the one of Table 2 and no richer. The reasons are those of the first report: a model asked for TEI writes it fluently, and fluency is the hazard, since well-formed markup with an invented attribute has no structural signature; an eight-macro language has a small surface and each macro answers one editorial question; and because the TEI is derived, a difference between a file and its TEI is a bug in the converter and not a judgement. The LaTeX with eight private macros is legible only to this repository. TEI is what an archive, a deposit or another editor can take in without the edition's rendering chain, and a transcription that can be cited only through this site dies with it.

The header carries what the file's comment carries, in a form a machine can read. The responsibility statement names the model as a `<name type="model">` with an identifier every editorial note points to, and dates the pass; the edition statement repeats the watermark; the availability is restricted and says why — the fonds is under copyright, the transcription is an unauthorised demonstration, no image is reproduced; the manuscript description carries the shelfmark, the folder's title as the inventory gives it, and the inventory's dating in an `<origDate>` with a note that the brackets are the archivists'; and the surrogate is Montpellier's

PDF, with a note explaining that each page break's @facs points one page ahead of its @n, because of the generated cover.

6.2 Mathematics as TeX

Mathematics is carried as TeX, untouched, inside `<formula notation="TeX">`, displayed formulas marked with `@rend`, and a commutative diagram as `<figure type="diagram">` holding a `<formula notation="tikz-cd">`. The few drawings on which an argument rests and which are not arrow graphs — arcs, chains of faces, pseudo-lines — are redrawn in TikZ and carried as `<figure type="picture">`, holding the source in a `<formula notation="TikZ">` and, in a `<graphic>`, the SVG compiled from it once, since no browser draws TikZ. TEI does not try to encode the mathematics, and neither does the converter. There are three reasons, and one cost.

The first is that the transcription is TeX already, and TeX is closer to what he wrote than MathML would be. He wrote a notation, not a structure: an underlined ω , a doubly underlined Σ , a tilde that distinguishes two coverings, a letter he used for two objects on one page. A presentation MathML of these would be a second transcription of the first, and a content MathML would be an interpretation — which is the modernised reading's work, and belongs in that register. The second is the apparatus. An illegible exponent or a doubtful index is an apparatus mark inside a formula, and the transcription writes it there, as `\ill{}` inside `$.;$`; kept as TeX, it stays where both the browser's renderer (KaTeX) and the PDF read it. A conversion to MathML would have to decide what an illegible subscript is, structurally, and there is no honest answer. The third is the round trip: the TeX inside the TEI is the TeX of the source, character for character, so the TEI can be checked against the `.tex` by comparing strings.

The cost is that the TEI says nothing about the mathematics that a TEI processor can use. A reader of the TEI can find every formula and cannot search inside one; the 1,395 commutative diagrams are figures whose content is opaque to the schema. And an apparatus mark inside a formula is invisible to TEI: a count of `<gap>` elements over the corpus misses every illegible index, which is why the edition's own counts (Section 7.3) are taken from the source and not from the TEI. The choice is recorded as a choice. MathML can be derived later from the same TeX, where the TeX is clean enough to support it, without touching the transcription. This article, encoded against the journal's own customisation, meets the question from the other side: that customisation admits no `<formula>`, and the article's TEI carries its few inline symbols as Unicode.

6.3 Validation, and what is not yet validated

Until late September 2026 each TEI file was checked for well-formedness as it was written, where the checker was installed, and nothing more ran on the build; validation against `tei_all` had last been run by hand on 5 September, over the 53 files that then existed. The edition now has a customisation of its own, on the model of its sibling on the Hopper ledgers, which learned the cost of going without one the hard way — a corpus of well-formed files that were all invalid, because nothing had asked (Hua 2026). The ODD declares exactly the elements the converter emits: sixty-four, header included, which are the sixty-two it emitted before, `<add>` since his insertions were told apart from the transcriber's (Section 4.4), and the `<schemaRef>` by which each file names the ODD. It closes the value lists of the attributes that carry meaning:

@type on <div>, <figure> and <note>, @notation on <formula> (TeX, tikz-cd), @reason on <gap>, and @rend wherever the converter writes it. It is a pure subset, so a file valid against it is valid TEI.

The RELAX NG schema is derived from the ODD on every build, with the Consortium's own stylesheets, and published beside it; every exported file names it in an `xml-model` processing instruction, so that an XML editor opening a downloaded transcription validates it unasked. The build then validates every file, with jing, against unmodified `tei_all` and against the derived schema, every input pinned by version and digest; and it reads the ODD against the export in both directions, failing on an element used and not declared and on one declared and never used, since a subset that can only grow is the full schema again. The first run found all 363 files valid against `tei_all`, and all of them invalid against the first draft of the ODD, in the same way: the two notes the header writes about the dating and the facsimile carry no @type. They belong to the edition and not to the apparatus, and the ODD now says that the attribute is absent there and only there, which the validator checks. A comparison of the TEI's reading view with the .tex's, page by page, with every formula of both typeset by KaTeX, runs on the same build and found no difference. What is still not validated is the Schematron that `tei_all` embeds, which is not run over the corpus, and the TeX inside each <formula>, which the schema sees as a string and only KaTeX reads.

The header's debt is paid in the same change. The export used to record the first pass and no other, so that the twenty-seven files carrying a later machine revision named one model and one date for a file two passes produced. Every pass the source's header records is now a <change> in <revisionDesc>, dated and pointing to the model that made it, with the source's own account of what changed; a model that revised a file without having first read it gets a responsibility statement of its own. Five files wrote a pass in a form no parser could read, one of them a first pass; they were rewritten at source, from the evidence of the file and its commits, and a record the converter cannot read is now refused rather than dropped.

7 Interface and reproducibility

7.1 A facsimile the edition does not hold

The site at <https://grothendieck.commutator.io> renders the transcription in one pane and Montpellier's facsimile in the other, and scrolling the transcription past a page break turns the facsimile to that page. The address of a reading is the page, not the batch: #140-1/2/p28 opens folder 140-1 at batch 2 with the transcription scrolled to page 28 and the facsimile turned to it, and every register below links in that form. The edition holds no page of the fonds: not in the repository, whose ignore file refuses PDF outright, and not on the server. The facsimile is Montpellier's own file.

Pointing an embedded frame at Montpellier's server does not work, for three reasons measured and recorded in the repository: the server sends `X-Frame-Options: SAMEORIGIN`, which forbids any other site from framing its files; its certificate had expired on 10 December 2025, so a framed load fails verification and the warning cannot render inside a frame; and it sends no cross-origin headers, so a script cannot fetch the file either. All three are enforced by the browser against the remote origin, and none applies to a request made by a server. So a relay on the edition's origin requests the file and passes the bytes through, verified byte-identical, forwarding range requests: opening page 221 of the 204-megabyte second volume of the Long March moves 256 kilobytes in a fifth of a second. The relay is a small Node service because

skipping verification of an expired certificate needs an option that no edge runtime offers; it serves only an allowlist of the folders' own files, which the build checks against the inventory; and it stores nothing.

This arrangement is an editorial fact and not only a technical one. The reading can be checked against its witness only while the witness answers. When Montpellier is down, the pane says so rather than offering a copy, and a rule of the repository forbids serving the local mirror the transcription passes read from, even in development. A reading beside an image the edition holds is a reading beside a copy; a reading beside the institution's own file is a reading beside the witness, and the difference is what a citation of it means.

7.2 Observed, not declared

Everything the site shows beyond the transcription is derived from it, and the counts of what exists are observed from the files and never kept by hand. The manifest of what is transcribed is built from the files present; a batch counts as transcribed once its LaTeX exists and as modernised once the folder's reading does, and neither claims that anyone has checked it. The site's progress figure is computed against the folders nobody has edited, and prints beside it the figure against the whole fonds, because narrowing a denominator flatters a ratio for free.

7.3 Registers derived from the reading

Once the transcription is a constrained language rather than a document, a question put to the fonds becomes a script. The edition publishes seven such registers, each regenerated from the files and each resting, entry by entry, on a file and a line.

The hand counts the apparatus per batch and per folder: of 848,095 words read by 3 October 2026, 64,030 were left illegible, 7.5 per cent; 45,286 readings were marked doubtful; he struck 36,306 times and inserted 13,069 times outside the transcriber's notes, the transcriber supplied 690 letters or words, and there are 5,342 marginal notes; there are 1,832 commutative diagrams and 847 drawings, described in notes and, for four of them, redrawn. Figure 3 places each folder's share of unread words against its inventory dating. The figure is a count of the apparatus and not of the ink, and says so in its caption: an `\ill{}` is a word one pass could not read, a fact about the hand and the pass at once.

The timeline draws each folder's inventory dating as a range — dark where the archivists read it, light where they deduced it, fading where it is open-ended — and the dates on the leaves as dots over it (Figure 1 is its summary); the parse of a dating keeps each end's qualifier, «vers», «à partir de», «avant», because collapsing «1965-[vers 1971]» to 1965–1971 would throw away the distinction the page exists to show. *The letters* is a register, not a text: 76 letters in 33 folders, 1959 to 1991, each with its date as written, its direction, its form, and how the transcription gives it, private correspondents unnamed. *The index* leads from a person, a work of his cited by volume (EGA, SGA, FGA), or a subject, to the folder and batch where it occurs. *The maps* are co-occurrence networks in the manner of bibliometric mapping — who is named together in a folder, which subjects share folders, which folders cite which volumes — whose every edge is evidence and whose layout is computed once, with a fixed seed, so that the figure is the same on every load. A slider over the years shows each map as it stood when the folders dated by then are counted — by the midpoint of the inventory's dating, which the control says is a tendency and not a date.

The last two registers are the mathematics itself, gathered out of the pages it stands on. *The diagrams* are the 1,832 `tikzcd` environments of the transcriptions, in 138 folders, shown one

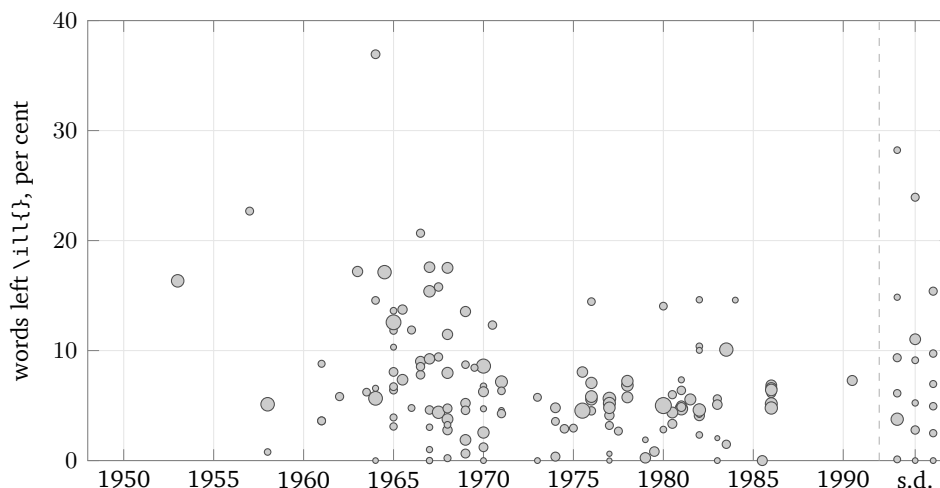


Figure 3: The share of words the first pass could not read, folder by folder, against the middle of the inventory's dating; 18 undated folders stand in the column at the right. A larger dot, more pages transcribed. 160 folders, 848,095 words read and 64,030 left illegible (7.0 per cent). No trend is drawn: the datings are ranges, often decades wide. The count is of the apparatus, not of the ink. From `src/content/hand.json` (2026-10-03), after the site's page *The hand*.

to a slide by the reading view's own renderer; *the formulas* are the 22,867 displayed formulas, in 152 folders, set by KaTeX with their apparatus inside. Each opens on its richest, ranked by a count that says what it measures: the distinct symbols a diagram or a formula uses — a letter in a given alphabet, so that $\mathcal{C}$ is not C , a named symbol, an operator — with, for a diagram, its nodes and arrows, and, for a formula, its size. The ranking finds what carries the most, not what matters most, which no count can find; and the richest diagram of the fonds, nineteen objects and twenty-nine arrows on page 28 of folder 140-1, a draft of the *Longue Marche*, is a fair sample of what the transcriptions ask a renderer to draw. Gathering them exposed faults a page of prose had hidden: every diagram of a page took its arrowheads from the first diagram's definitions, so that a hidden first diagram left every arrow headless; a label was set a fixed distance from its arrow whatever its size, and a long one sat on the line; and KaTeX's stretchy braces, painted at a negative depth, vanished under any background. None was a fault of the transcriptions, and all were found by looking at the mathematics one piece at a time rather than a page at a time.

Two rules keep these registers honest. The first is that the transcriber's notes are excluded from every count of his words, since a note that discusses a notion would otherwise report it as his. The second concerns who may appear as a node. A person counts where the folder holds a letter to or from them, a text of theirs, or his naming them in words that make it an attribution — «th. de Hodge», «d'après Deligne» — and not where an object merely bears a name, «groupe de Galois»; the test is adjacency in the text, and an earlier, looser test let «espace de Hilbert» through. And a source judged not reliable is not used, even where it is the only source for a claim: the claim is then unsourced, and the graph goes without it.

8 Findings, kept from becoming claims

Reading a folder end to end turns things up that the inventory does not say: a leaf bound backwards, two manuscripts interleaved, a theorem proved under a hypothesis weaker than the one in print. The edition keeps them in a list, and the list is built around refusing to

overclaim, because a claim that something in these folders is new is a claim about the literature, which can be searched but never exhausted, and a claim that it came first needs a date, which the folders mostly lack.

Each row carries a status from a closed list: *unsearched*, nobody looked it up; *candidate*, searched and not found; *matched*, found in the literature and kept as a killed candidate; *refuted*, shown false as stated by a counterexample, kept with what of it still stands; *confirmed*, which only a person may set. Every row names the sources searched and separates what the page carries from what the edition supplied. The skill that proposes rows reads the transcription and the modernised reading, never the facsimile, and may not assert priority. Killed rows stay on purpose: a matched candidate saves the next reader the search, and a list that only grows is not being checked. On 3 October 2026 the list held 143 rows from 41 folders: 56 unsearched, 71 candidates, 15 matched, one refuted, and none confirmed.

One row shows what the list is for. Folder 27 holds five typed leaves headed in pencil «Exp XIII», a proposition on a group acting on a scheme with a chain of implications among six conditions and three corollaries. A first pass summarised them as a typescript already published. A comparison with the exposé in the current re-edition of SGA 3 (Gille and Polo 2008–2011, Exposé XIII in its version of December 2008) found the premise wrong: the published exposé goes from its first lemma straight to its second section, and the general statements reappear there only specialised. The leaves are an earlier, longer version not in print, and were retranscribed in full. Nothing in this establishes that the version matters; it establishes that the inventory's «typescript» and the literature's «published» were not the same text, which a reader of the folder could not have known.

9 Discussion

9.1 The reading against the image

The transcription is a first pass and no batch has been checked by a person. On 27 September 2026 the editor checked some fifty points against the facsimile: twenty readings — dates, doubtful words, whose hand an addition is in —, the ink of nineteen typed batches, sampled on one page each, and eleven untranscribed pages, found blank; each is recorded, dated and signed, in the file where it was made, and none makes its batch checked. In October the editor settled on the facsimile the sixty-three points at which ten documents differ from a transcription made independently elsewhere — six letters and four pieces — and four passages the pass had left illegible; twenty-three readings were corrected, and the rest kept as transcribed. On 3 October the page of folder 19 shown in Section 4.5 was read against the facsimile in the same way, and two of its readings corrected (Appendix A). The edition's status file, in which only a person may declare a batch checked or skipped, declares no batch checked and two skipped; the site shows every batch as drafted because that is what the files prove. The rate at which the reading fails is stated in the apparatus and nowhere else: 7.5 per cent of words left illegible, a further 5.3 readings in a hundred words marked doubtful, varying by folder from nothing to more than a third (Figure 3). Those rates are not error rates. A pass that read a word wrongly and plainly left no mark, and the only instrument that finds such a word is a person with the facsimile.

9.2 Limits of the model as a transcriber and as an encoder

Five limits, none of them an error rate.

One macro carried two senses. `\add` was declared as the transcriber's addition — a missing letter, an implied word — and mapped to `<supplied resp="#pass">`. Read in context, most of its 6,144 uses were his: a word written above the line with a caret, a substitution over a struck word, the α_i of Section 4.5. Sixteen batch headers stated as much for their typescripts, where his ink insertions were marked with it by convention. The TEI credited to the model several thousand of his insertions, and a count of them from the TEI was wrong by an order of magnitude. The same failure was found in the Hopper edition, on a smaller scale, and resolved there by dividing the macro in two; it has been resolved here the same way. `\supplied` now carries the transcriber's sense and maps to `<supplied resp="#pass">`; `\add` carries his and maps to `<add>`. The corpus was divided between them by a rule checkable in the file rather than by a fresh reading: an addition glued to the word it completes, a letter or digit immediately before or after it, is the transcriber's, and one standing free is his. That assigns 529 uses to `\supplied` and leaves 5,615 as `\add`. The residue is the rule's guess and not a reading. A free-standing supply — his sign « = » for « même » expanded in full, a closing parenthesis restored at a clipped edge — still reads as his, and a prefix he inserted before a word, now « `\supplied{quasi-}torseur` », has been handed to the transcriber. Only a person reading the facsimiles will settle those. Given one macro for an insertion, the model used it for both kinds, and the lesson generalises: a macro whose declared sense is narrower than its natural sense will be stretched to the natural one.

The model knows the mathematics. Its prior is the hazard the rule of Section 4.3 is written against, and the rule holds it only where it is applied. An illegible index inside a formula that the context determines is a standing temptation; the transcriptions resolve it by writing `\i11{}` and, where the context fixes the value, saying so in a note — « l'indice de k est un pâté ... le contexte ne le fixe pas » — but a pass that supplied the index silently would leave no trace. The separation of registers is the method's response: completion belongs to the modernised reading, which is allowed to say what is true, and is forbidden to the transcription, which is not.

The apparatus is inside the mathematics. Section 6.2 set out why TeX is kept; the cost to the apparatus is that a count over the TEI misses every mark inside a formula. The edition's counts are taken from the source for that reason. A consumer of the TEI alone would undercount the illegible by an amount the edition can state and the TEI cannot.

Other people's texts. The fonds holds letters, typescripts and mémoires by others, some of whom are private persons. The rules of Section 4.3 answer to rights and not to the model, and they have to be applied by the model: whether a writer is a public figure is a judgement the skill makes conservatively and the register of letters records, with fourteen of its seventy-six records leaving the correspondent unnamed. A human reviewer is the only check on that judgement, and its failure is not a misreading but a publication.

Dates. The model reads a date as readily as a word, and a date is the one reading the edition's own apparatus most wants: the inventory's datings are mostly inferred (Figure 1). The register of dated leaves distinguishes a date in his hand from a typist's and a machine's, keeps two-digit years as written, and fixes a year from a neighbouring leaf only where the run is continuous, saying so. Its 160 records were assembled by a model reading the transcriptions and were not checked against the facsimile.

9.3 Human validation, and the pace

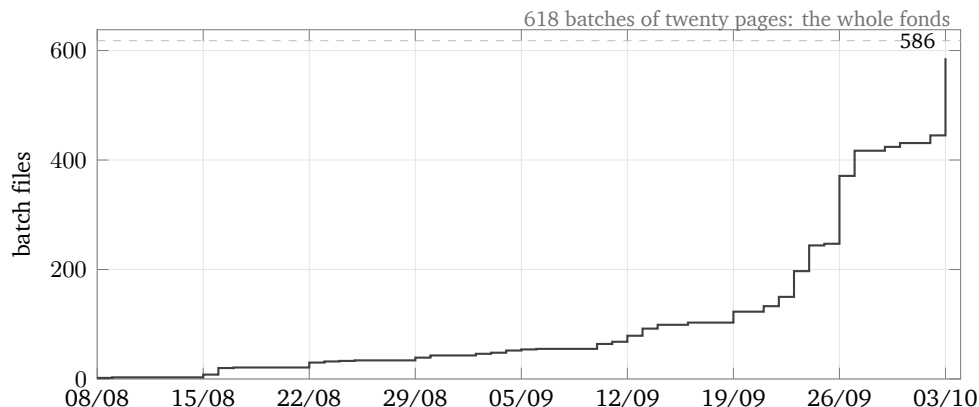


Figure 4: Batch files in the repository, day by day, from the first on 2026-08-08 to 2026-10-03: the date each batch-`NN.fr.tex` was first committed, cumulated. Each file is one pass of the model over at most twenty pages. The dashed line is the whole fonds at that rate. From the repository's git history, drawn on 2026-10-03 by `docs/study/figures.mjs`.

Figure 4 gives the pace from the repository's history: the repository was opened on 7 August 2026, the first batch file was committed the next day and the 586th on 3 October; 43 were committed in August, 388 in September and 155 in the first three days of October, 141 of them on 3 October alone, with up to eight passes running at once. At that rate the part of the fonds left to this edition is weeks of machine time. The human check is not. A batch of twenty pages of this hand, compared line by line with the facsimile by a mathematician who can read it, is an afternoon; the edition's 618 batches are some two and a half thousand hours of a rare kind of reader. That is the size of what is left, and it is the argument of this article in one number: the transcription is no longer the bottleneck and the checking is, and an edition that cannot say which of its pages a person has read is publishing a machine's confidence as though it were scholarship.

The protocol the edition proposes is the one its status file already encodes. A person opens a batch beside its facsimile, reads each transcribed line against the page, corrects the transcription at source with a note saying who checked it, and only then may declare the batch checked; the citation of every page in it then says so. Until then every citation names a machine pass and a date, and the address of the page does not move when the reading does.

9.4 What the method permits, and what it does not

The method permits a reading of a sixteen-thousand-page mathematical archive to be produced in months by one person's attention and a model's time, cited by folder, batch, page and pass, exported to TEI with its mathematics intact, placed beside an institution's own facsimile without copying it, and mined for registers of names, dates, letters and hands that each rest on a line. It does not produce a checked transcription; it does not encode the mathematics in a form a TEI processor can reason over, though it validates what it does encode against a customisation of its own; it does not decide who is a public figure with authority; and it does not remove the need for the person it was designed around.

10 Conclusion

This article has described a second edition built on the method of the first, and what a corpus of mathematics changes in it. The notation required a decision the ledgers did not: to carry mathematics as the TeX it was transcribed in, inside `<formula>`, rather than as a structure the transcription does not have, and to put the interpretation of that notation in a second register derived from the first. The size of the corpus turned the human check from a stated debt into the edition's main fact. And the rights turned the facsimile from a file the edition serves into one it points at, through a relay that holds nothing, so that the reading and its witness meet only on the reader's screen.

The encoding work this article first listed as owed is done: an ODD declaring exactly what the converter emits and closing its value lists, a schema derived from it that every file names, validation of every file against it and against `tei_all` on every build, a header that records every pass, and a refused file that stops the build, so that a deposit is TEI and not XML. What remains is stated as two pieces of work. The first is the reading against the facsimile of what the rule of Section 9.2 divided mechanically: which of the 5,615 free-standing insertions are his and which of the 529 glued ones are the transcriber's. The second is the human check, which is the one on which every claim of the edition depends, and which is a question for the mathematicians who can read this hand, not for a model.

An edition is worth what its reader can check. For this fonds, checking means reading Grothendieck's hand against his page, which few people can do, over sixteen thousand pages, which nobody can do alone. What the method changes is that the reading exists, fully cited, before anyone checks it; what it cannot change is who must.

About the author

Michel Hua is an independent consultant at Commutator, Paris, working with large organisations on data and cloud applications: platform migrations, data governance, and the adoption of generative AI. The editions he has built apply that discipline to archives institutions have digitised and few have read: the Hopper ledgers at the Whitney Museum, the Grothendieck fonds at Montpellier, and the mathematicians' manuscripts in Gallica. michel@commutator.io.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude Opus 5, Claude Opus 5.5, Claude Fable 5 and Claude Fable 5.1 (Anthropic), through Claude Code, in order to transcribe the pages of the fonds into the restricted transcription language described in Section 4, to write the modernised readings described in Section 5, to write the scripts, the converter, the relay, the interface and the derived registers described in Sections 6 and 7, to propose the findings described in Section 8, and to draft this article from the transcriptions and the derived data under the author's direction. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. The transcriptions have not been checked against the facsimiles by a person, and the article says so wherever a claim depends on them.

Data availability

The transcriptions, the modernised readings, the TEI export, the scripts, the derived registers, the relay, the interface and the sources of this article are in the public repository at <https://github.com/Commutator-I0/grothendieck-project>; the interface is served at <https://grothendieck.commutator.io>. The software and the derived data are released under the repository's licence; this article under CC BY 4.0. The fonds is the University of Montpellier's and is under copyright; the transcriptions are unauthorised working documents, published for study and removed on request, and no image of the fonds is stored by the edition or reproduced in this article.

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A A complete transcribed page

Page 8 of folder 19, batch 1, as the model wrote it in the restricted language, from the first pass of 22 August 2026 by Claude Opus 5; the TEI of Section 4.5 is derived from it. The page is on the site, beside Montpellier’s facsimile.

```
\page{8}
qui s'expliciteront en termes des  $\varphi_j = f_j \circ g_i$ , et des
 $\bar{\alpha}_i$ ,  $\bar{\lambda}_i$  correspondants, ainsi
\l[
\left\{
\begin{array}{l}
\bar{\alpha}_k = \alpha_k(X_k) \circ \mathrm{pr}_k \llbracket 2pt \\
\bar{\lambda}_{k,j,i} = \lambda_{kji}(X_i) \circ \mathrm{pr}_i
\end{array}
\right.
\tag{3.9}
\l]
\l[
\lambda_{kji} : \varphi_{ki} \longrightarrow \varphi_{kj} \circ \varphi_{ji}
\tag{3.10}
\l]

\begin{tikzcd}[column sep=small, row sep=small, nodes={font=\scriptsize}]
\varphi_{ki} \arrow[r, "\lambda_{kji}"] \arrow[d, no head] & \varphi_{kj} \varphi_{ji} \arrow[d, no head] \\
f_k \circ g_i & f_k \circ g_j \circ f_j \circ g_i
\end{tikzcd}

\note{la flèche provient du morphisme d'adjonction  $\mathrm{id} \rightarrow g_j \circ f_j$ ,
que l'auteur indique par une flèche verticale sous  $g_j \circ f_j$ }
```

Les $(\varphi_{ji}, \lambda_{kji})$ sont reliées par des conditions exprimant (1.5), (1.6), que nous n'expliciterons pas.

\marginal{pour $i = j = k$, λ_{kji} n'est autre que λ_i }

[Le cas le plus intéressant est celui où les φ α_i sont des isomorphismes (i.e. les g_i pleinement fidèles). Dans ce cas, φ_{ij} on peut supposer $\varphi_i = \mathrm{id}_{B_i}$, $\alpha_i = \mathrm{id}_{\mathrm{id}_{B_i}}$, $\lambda_i = \mathrm{id}$.]

Les conditions (1.) φ dans les cas où deux des i, j, k sont égaux, avec $i = j$ ou $j = k$, φ ne montrent que l'un des λ_{kji} est l'identité, φ_{kj} φ_{ji} .

\note{ligne très surchargée ; le verbe reste douteux}

Il reste donc à examiner les cas (3.11) : a) i, j, k deux à deux différents ; b) $i = k \neq j$, d'où

\[\mathrm{id}_{B_i} \longrightarrow \varphi_{ij} \circ \varphi_{ji} \quad (i \neq j) \]

Read against the facsimile on 3 October 2026, the page needed two corrections. The two vertical strokes of the boxed diagram (3.10) are equality signs, which the pass had drawn as plain lines; and case a) of (3.11), « i, j, k tous différents», is not struck, while the pass had read it as struck and as «deux à deux différents». The file on the site carries both corrections and a Revised line naming them; the listing above is the first pass as it was, and the TEI of Section 4.5 is that of the corrected file.